ADR-02: Data Generation Strategy for Customer Query Analytics

Architecture Decision Record

September 8, 2025

1 Status

Accepted

2 Context

Customer query analytics requires high-quality annotated datasets to develop reliable classification systems and assess inter-annotator agreement. The project faces constraints around data availability, privacy regulations, and the need for controlled experimentation.

2.1 Data Requirements

- Realistic customer support ticket content across multiple difficulty levels
- Multiple annotator perspectives to assess inter-annotator agreement
- Hierarchical labels (L1: Technical/Billing/Account, L2: Specific subtypes)
- Sufficient volume for statistical analysis and pattern detection
- Diverse scenarios representing real-world annotation challenges

2.2 Alternative Approaches Considered

1. **Real Customer Data**: Use actual support tickets with anonymization
2. **Public Datasets**: Leverage existing customer service datasets
3. **Crowdsourced Annotation**: Hire external annotators for real data labeling
4. **Synthetic Data Generation**: AI-generated tickets with simulated annotator responses
5. **Hybrid Approach**: Combination of real data samples with synthetic augmentation

3 Decision

We selected **LLM Based Synthetic Data Generation with Multi-Annotator Simulation** using the following architecture:

3.1 Core Strategy

- Generate 50 hyper-realistic customer support tickets across three difficulty tiers
- Simulate 7 distinct annotator personas with varying skill levels and biases
- Implement controlled policy ambiguities to create realistic disagreement patterns
- Use tiered complexity (Easy: 15, Medium: 15, Hard: 20 samples) for comprehensive analysis

3.2 Annotator Simulation Framework

- **Tier 1 - Expert Annotators**: alex-001, beth-002, carlos-003 (high accuracy)
- **Tier 2 - Medium Annotators**: diane-004, eric-005 (policy-confused but well-intentioned)
- **Tier 3 - Poor Annotators**: fatima-006, george-007 (low-effort, keyword-based)

3.3 Complexity Stratification

- **Easy Tickets (15)**: Clear, unambiguous requests with high expected agreement
- **Medium Tickets (15)**: Policy ambiguities testing L1/L2 boundary decisions
- **Hard Tickets (20)**: Stream-of-consciousness, emotionally charged, multi-issue scenarios

4 Rationale

4.1 Advantages of Synthetic Approach

- **Privacy Compliance**: Zero risk of customer data exposure or regulatory violations
- **Controlled Experimentation**: Precisely engineered scenarios to test specific analytical capabilities
- **Annotator Bias Simulation**: Realistic human annotation patterns without human resource costs
- **Rapid Iteration**: Ability to generate additional samples or modify scenarios as needed
- **Ground Truth Control**: Known complexity levels enable validation of analytical methods

4.2 Technical Implementation Benefits

- **Consistent Quality**: Every ticket engineered to serve specific analytical purposes
- **Realistic Disagreement**: Annotator personas create authentic inter-annotator agreement patterns
- **Scalable Generation**: Framework can produce additional datasets for future analysis
- **Policy Testing**: Deliberate ambiguities validate annotation policy effectiveness

5 Consequences

5.1 Positive Outcomes

- **Risk Mitigation**: Eliminated data privacy and compliance concerns
- **Analytical Validity**: Controlled dataset enables rigorous testing of IAA methods
- **Cost Efficiency**: No annotation labor costs or real data acquisition expenses
- **Reproducibility**: Consistent dataset enables reliable algorithm comparison and validation
- **Educational Value**: Clear demonstration of annotation challenges and solution approaches

5.2 Trade-offs and Limitations

- **Synthetic Bias**: Generated content may miss nuances of real customer language
- **Limited Vocabulary**: Constrained to AI-generated linguistic patterns
- **Scale Constraints**: 50 samples sufficient for proof-of-concept but limited for production training
- **Domain Gaps**: May not capture industry-specific terminology or edge cases

5.3 Rejected Alternatives

- **Real Customer Data**: Legal and privacy risks outweighed analytical benefits
- **Public Datasets**: Available datasets lacked hierarchical structure and multi-annotator perspective
- **Crowdsourced Annotation**: High cost and quality control challenges
- **Hybrid Approach**: Complexity overhead without sufficient additional benefit

6 Implementation Details

6.1 Data Generation Parameters

- **Ticket Length**: 2-3 sentences (easy), 3-5 sentences (medium), 4-7 sentences (hard)
- **Emotional Range**: Neutral/polite (easy) to highly emotional/frustrated (hard)
- **Ambiguity Injection**: Systematic testing of Login vs. Password Reset boundary cases
- **Realism Features**: Typos, grammar errors, stream-of-consciousness patterns in hard tier

6.2 Quality Validation

- Manual review of generated samples for realism and diversity
- Verification that annotator personas produce expected disagreement patterns
- Confirmation that difficulty tiers create appropriate analytical challenges
- Cross-validation with subject matter experts on ticket authenticity

6.3 Success Metrics

- Inter-annotator agreement ranges: Easy ($\alpha > 0.8$), Medium ($0.4 < \alpha < 0.8$), Hard ($\alpha < 0.6$)
- Annotator performance differentiation clearly visible in confusion matrices
- Sufficient linguistic diversity for meaningful topic modeling and entity recognition
- Realistic sentiment distribution across emotional spectrum

7 Future Considerations

7.1 Evolution Path

- **Scale Expansion**: Generate larger datasets (500+ samples) for production model training
- **Real Data Integration**: Incorporate anonymized real samples for validation
- **Domain Extension**: Adapt generation framework for other business contexts
- **Dynamic Generation**: Real-time synthetic data creation for continuous testing

7.2 Validation Requirements

Quarterly assessment of synthetic data quality through:

- Comparison with emerging real customer ticket patterns
- Validation against production model performance
- Expert review of linguistic authenticity and business relevance
- Extension testing to additional customer service scenarios